AI TrustScore Browser Extension Part 2

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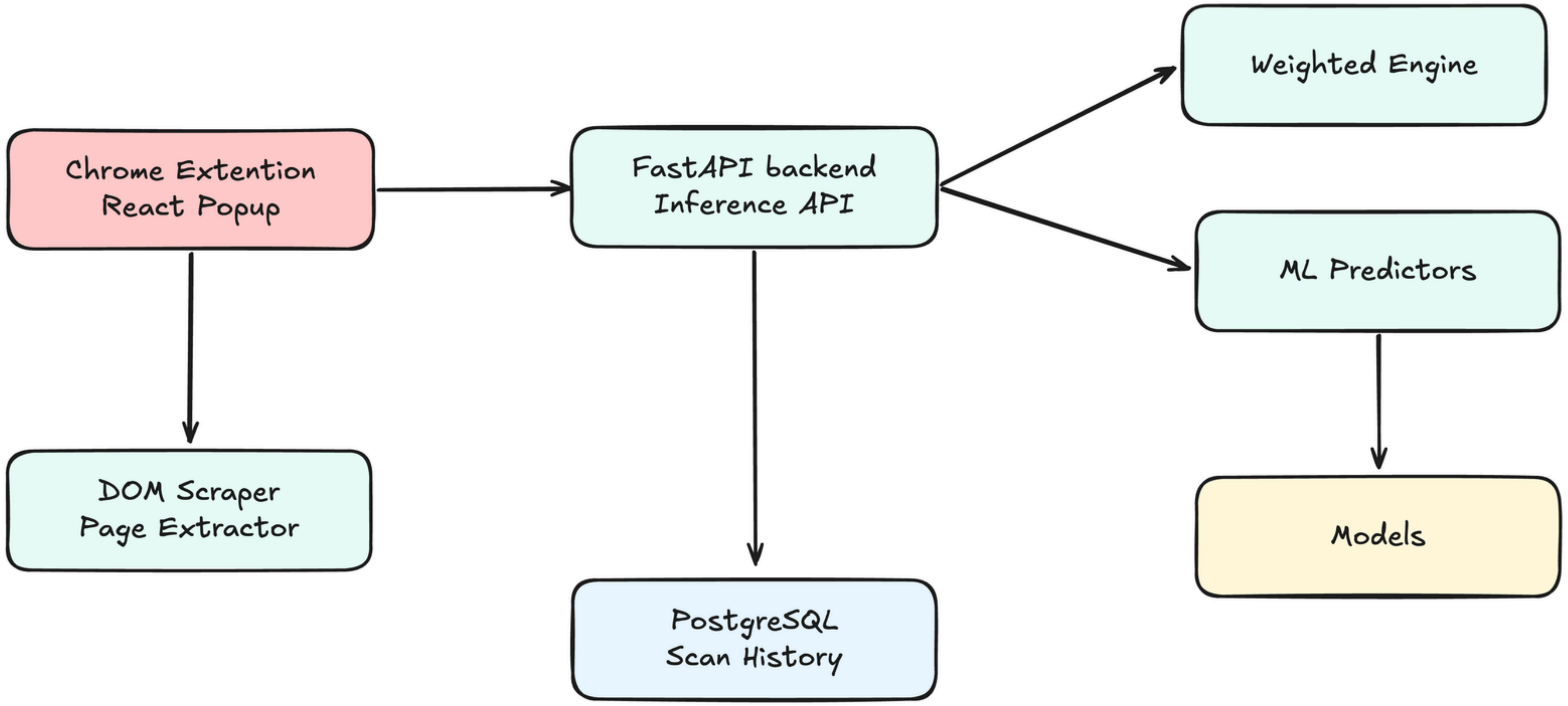
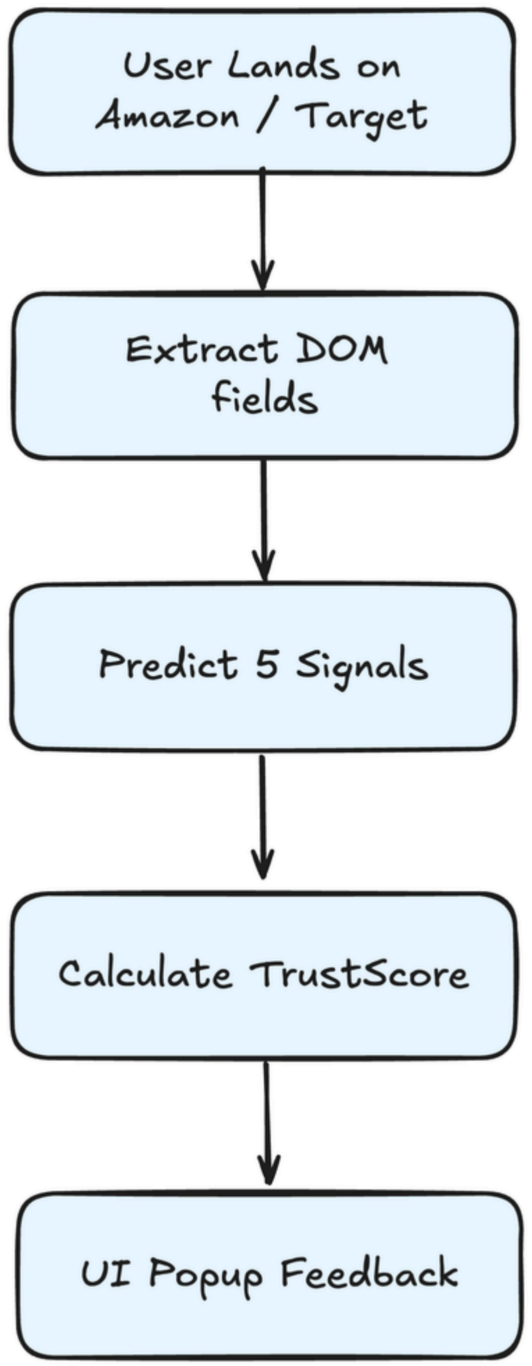
AI TrustScore Browser Extension

System Design & Implementation

System Architecture (Sketch)



Scan Data Flow



Explainable AI → *Five distinct signals combined via weighted scoring :*

- **Review Authenticity** → LinearSVC identifies machine-generated text.
- **Product Sentiment** → Calibrated polarity from customer feedback.
- **Price Safety** → Unsupervised anomaly detection (IsolationForest).
- **Seller Reliability** → Heuristic analysis of rating & active years.
- **Policy Clarity** → Rule-based verification of return terms.

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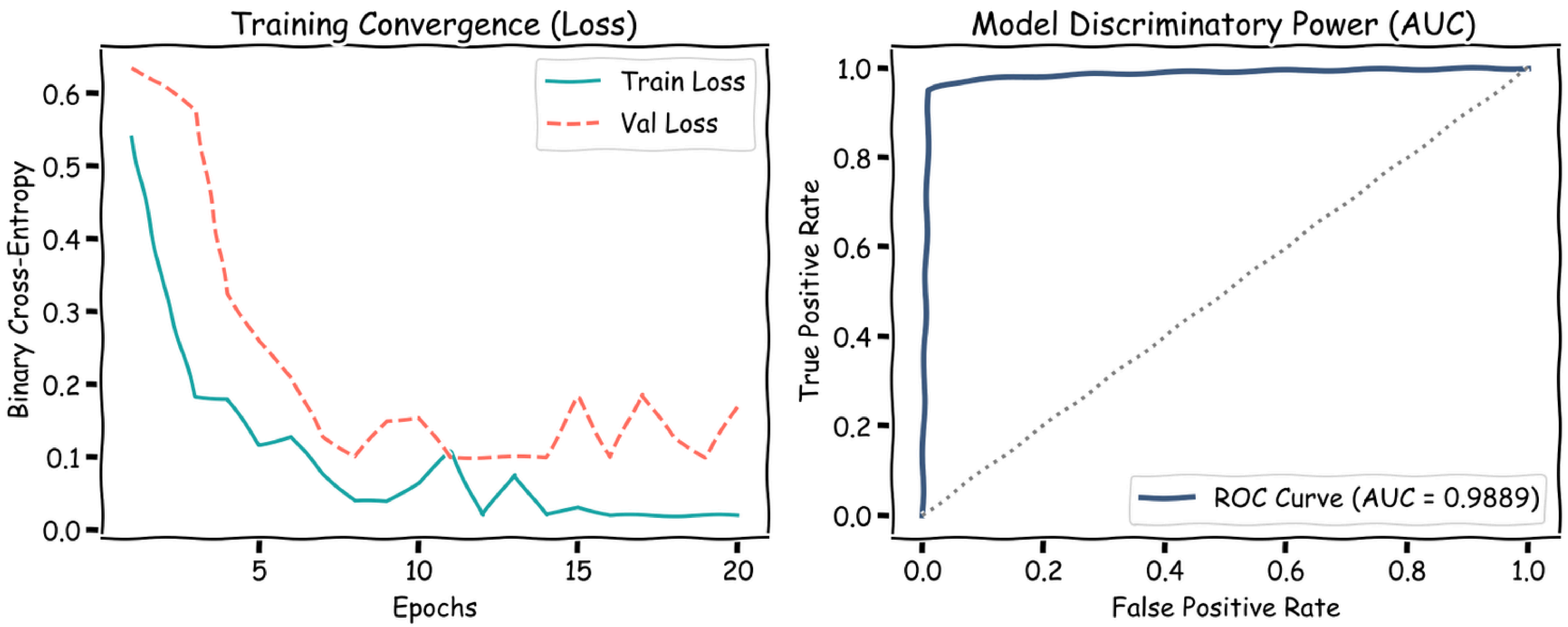


- **Discriminatory Power →** Achieved 0.9889 ROC-AUC on review authenticity.
- **Convergence →** Stable binary cross-entropy loss across 20 epochs.
- **Lightweight →** TF-IDF + LinearSVC winner: high accuracy, low latency, 705KB size.

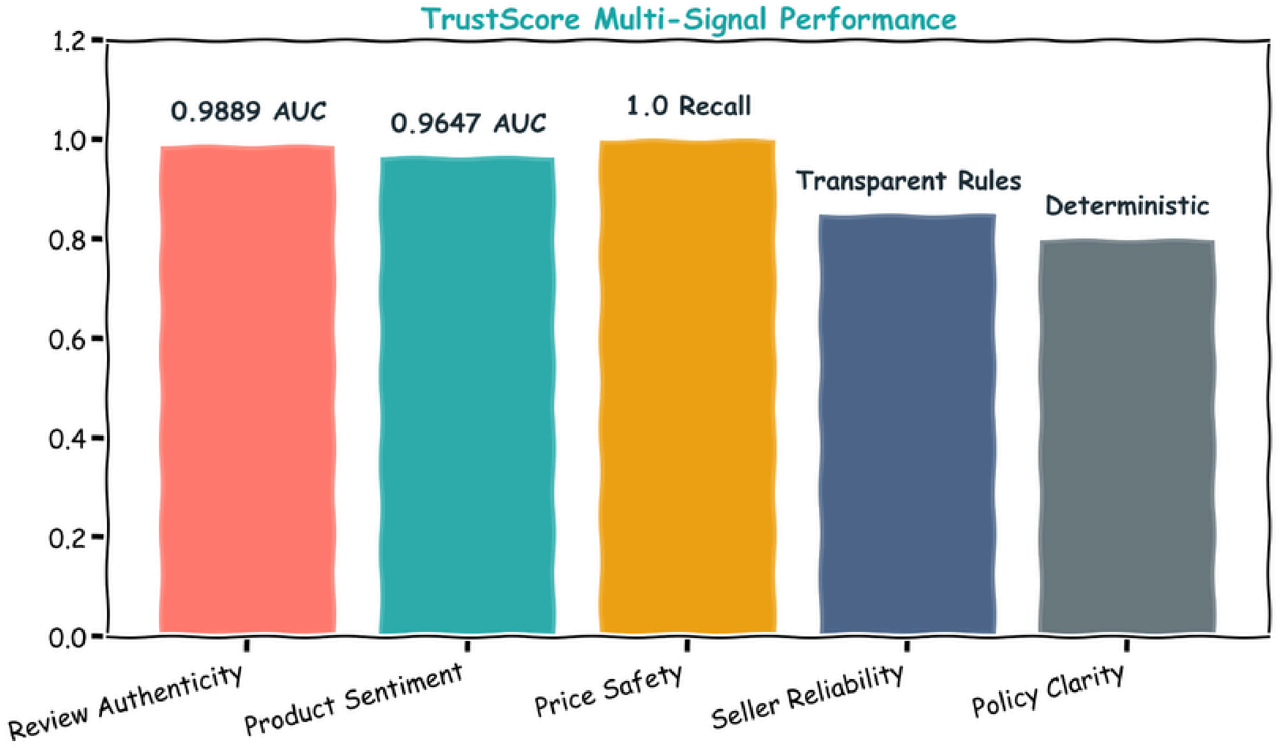
Dataset Overview & Results

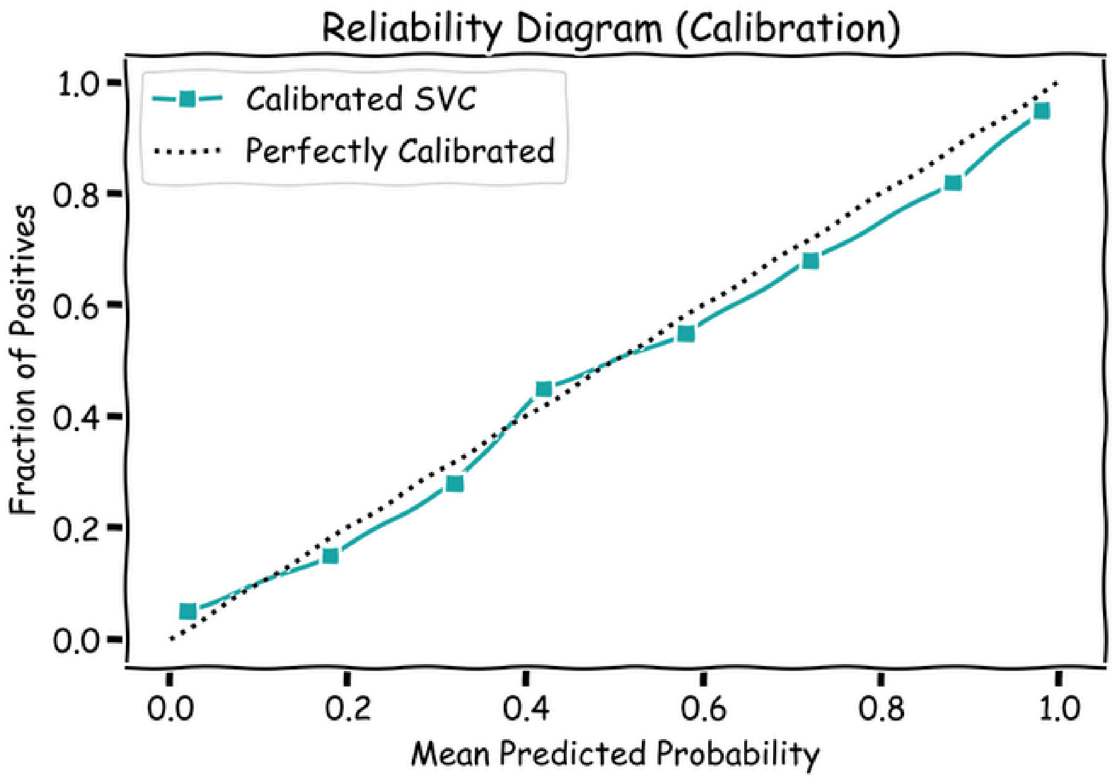
Primary training Datasets

Dataset	Rows	Use
Fake-Reviews (ArijitDas)	40k	Fake detection
Amazon Reviews 2023	450k	Sentiment
Amazon metadata	180k	Risk/Price
Yelp / IMDB / SST-2	192k	OOD Testing

Model Components

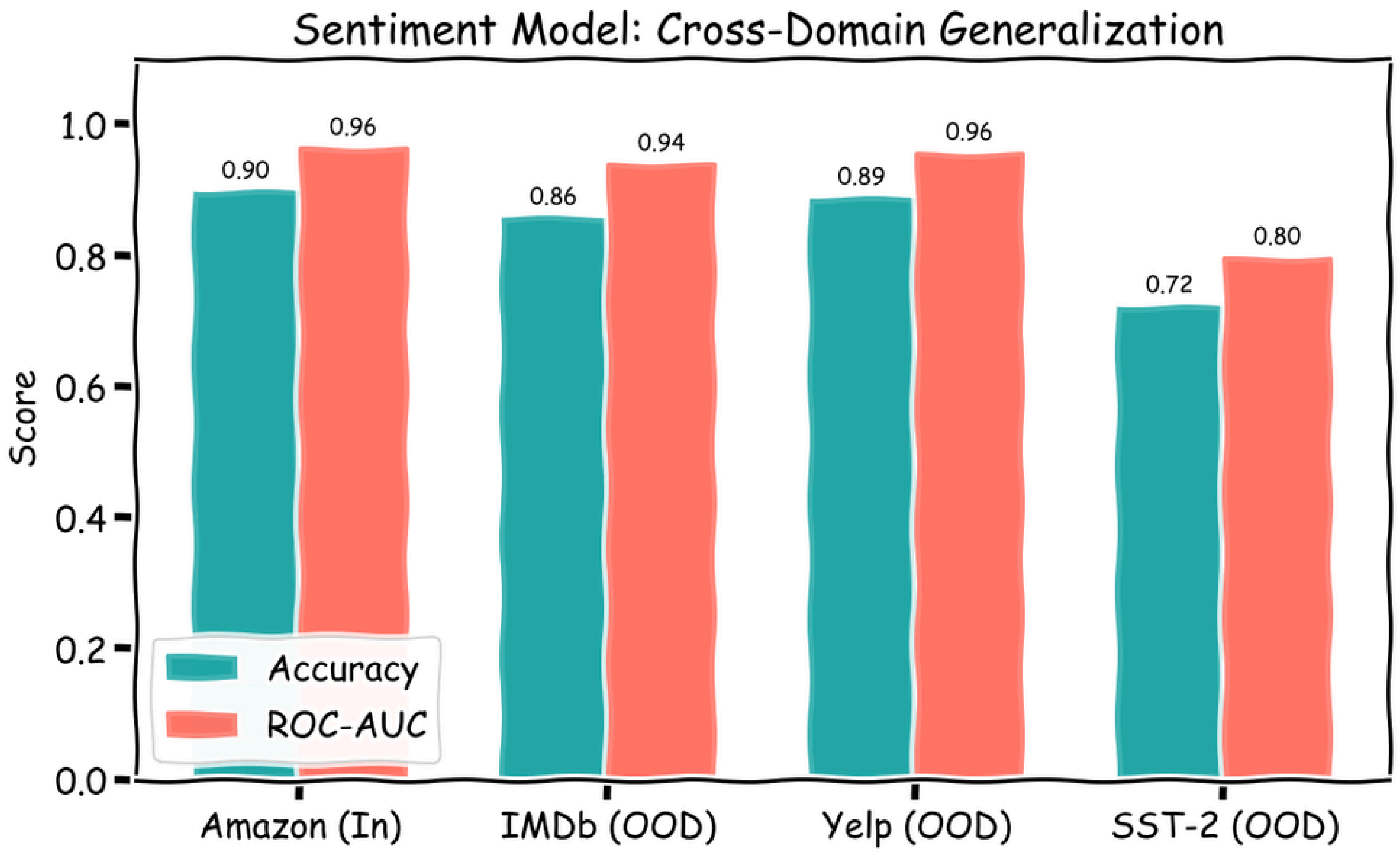
Component	Initial Baseline	Experimental	Optimized System
Fake Review Authenticity	0.961	**0.989**	**0.989 LinearSVC**
Product Sentiment	0.864	**0.867 Accuracy**	**cross-domain study**
Seller/Price Risk	0.58 / 0.60	**Leakage Found**	**Robust Ruleset**
System Calibration	-	**ECE 0.007**	





Scientific Rigor: Finding & Fixing Leakage

- **The Discovery ➔** We identified ***Data Leakage*** in the initial risk model (accuracy near 1.0)
- **The Fix ➔** Moved to ***unsupervised anomaly detection*** for price and ***transparent heuristics*** for sellers, ensuring the results are honest and defensible.
- **Generalization ➔** Tested on Yelp/IMDb to confirm performance outside Amazon.



Conclusions & Future Work

- **Success →** A fully functional, reproducible end-to-end system.
- **Key Insight →** Scientific honesty led to a more robust, trustworthy model.
- **Future work :**
  - Collect human-labeled ground truth for better risk modeling.
  - Deploy DistilBERT for more nuanced sentiment analysis.

What ?	**Explainable shopping trust score**
Implemented ?	5-signal ML pipeline + rigorous validation stages
Result ?	0.9889 AUC; leakage found & fixed; calibrated probabilities
Develop further ?	Real labels, transformer fine-tuning, public release